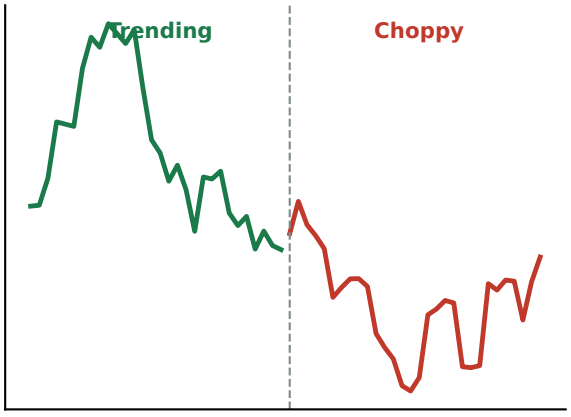
Regime Ensemble

Detecting market conditions to reduce risk — not just maximise returns

SPY · 2000–2025 · v2.0

What is a market regime?



Markets switch between trending and choppy phases.
Spot the switch early and you can avoid the worst of it.

Why does it matter?

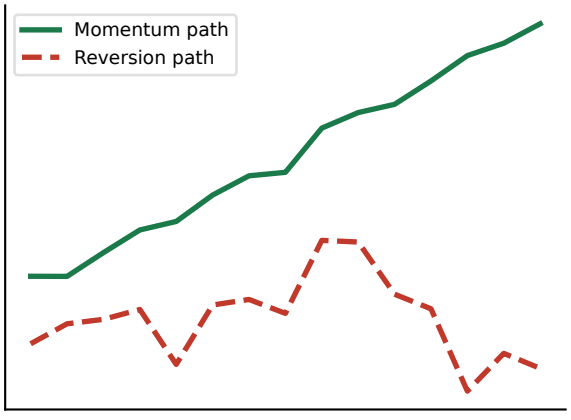
Stay invested when momentum is real,

step aside when conditions turn choppy.

The goal is not to beat the market every year —

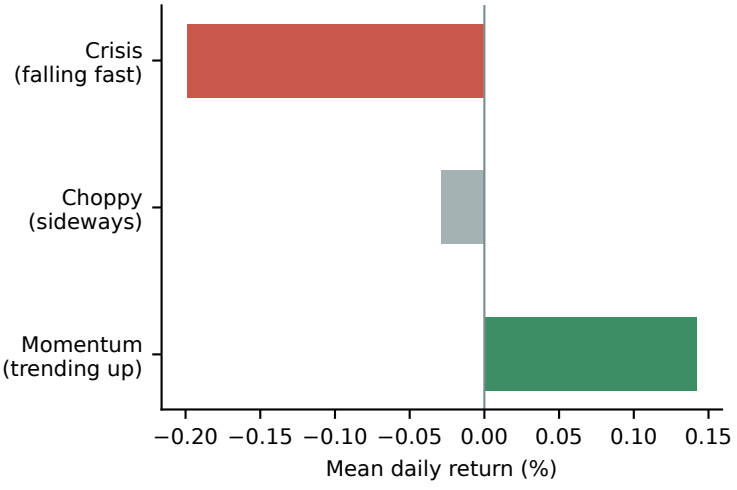
it's to avoid catastrophic drawdowns.

Detector 1 — Geometric (path shape)



Measures how straight the last 15 days of price movement was.
A straight path → trending. A zigzag → mean-reverting.

Detector 2 — Markov (hidden state)



A statistical model that learns 3 hidden market states from return patterns.
Identifies crisis conditions and suppresses buy signals automatically.

Headline Results (SPY 2000-2025, zero transaction costs)

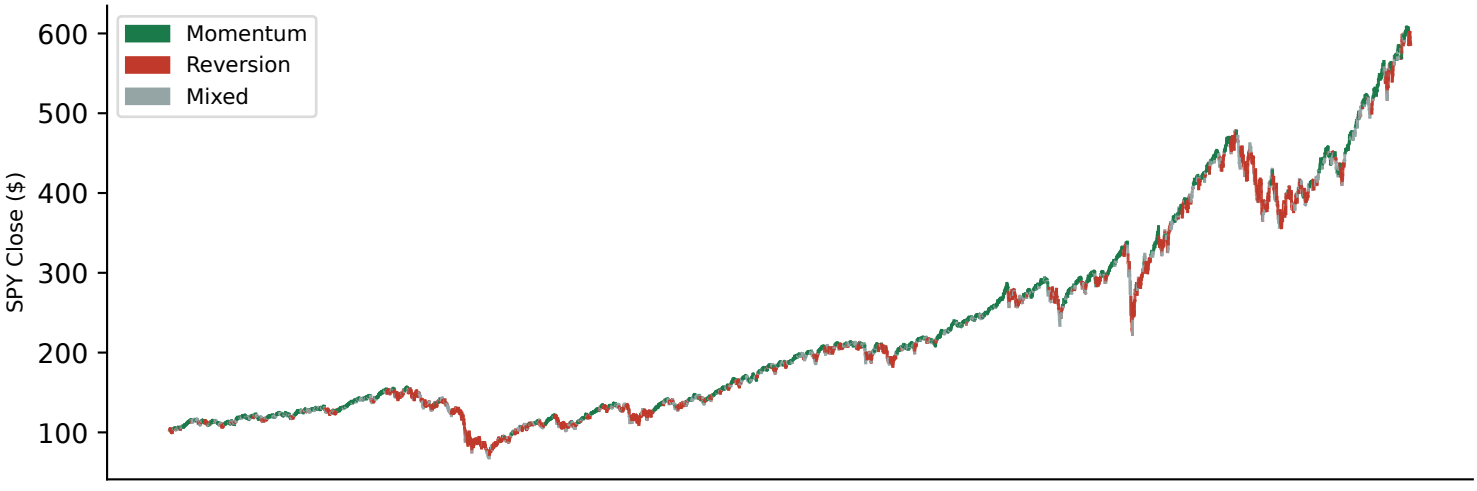
	CAGR	Sharpe Ratio	Max Drawdown	T-stat (momentum)
Strategy (Regime Ensemble)	**+1.8%**	**0.29**	**-15.9%**	**1.32**
Buy & Hold (benchmark)	+8.6%	0.44	-56.5%	2.02

⚠ At 10bps round-trip costs, Sharpe drops to 0.55. At 20bps, the strategy is unprofitable. Transaction costs are material — see page 3.

Models in Action — SPY 2000-2025 (Real Data)

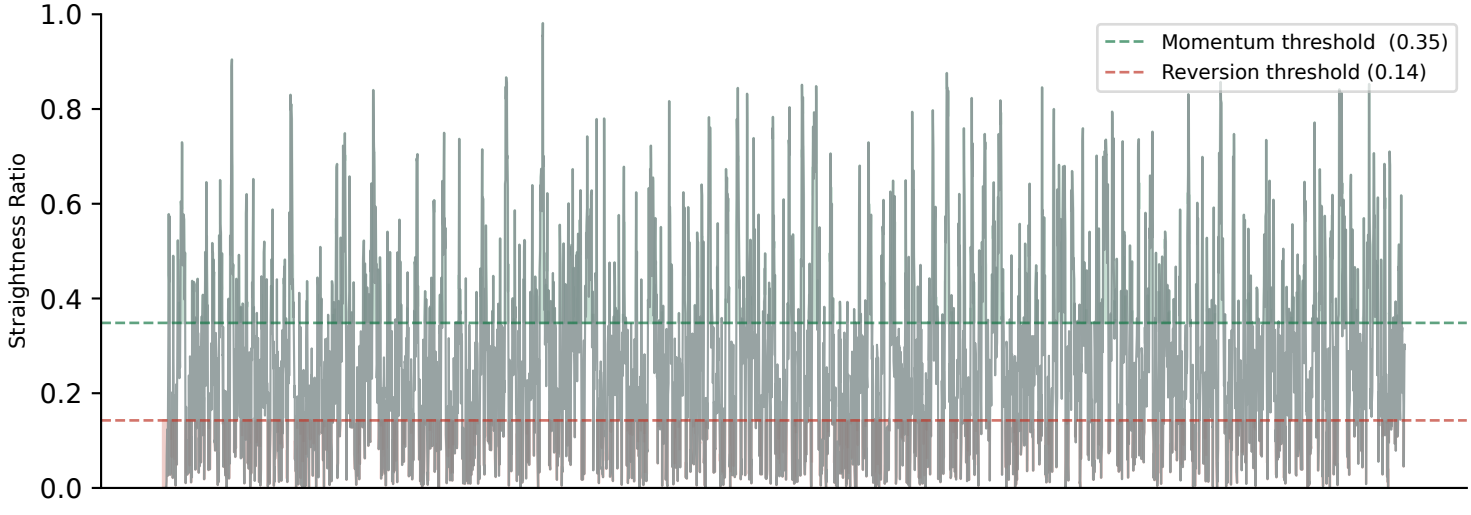
Three panels: price coloured by regime, geometric detector signal, Markov probabilities

SPY Price — Coloured by Ensemble Regime



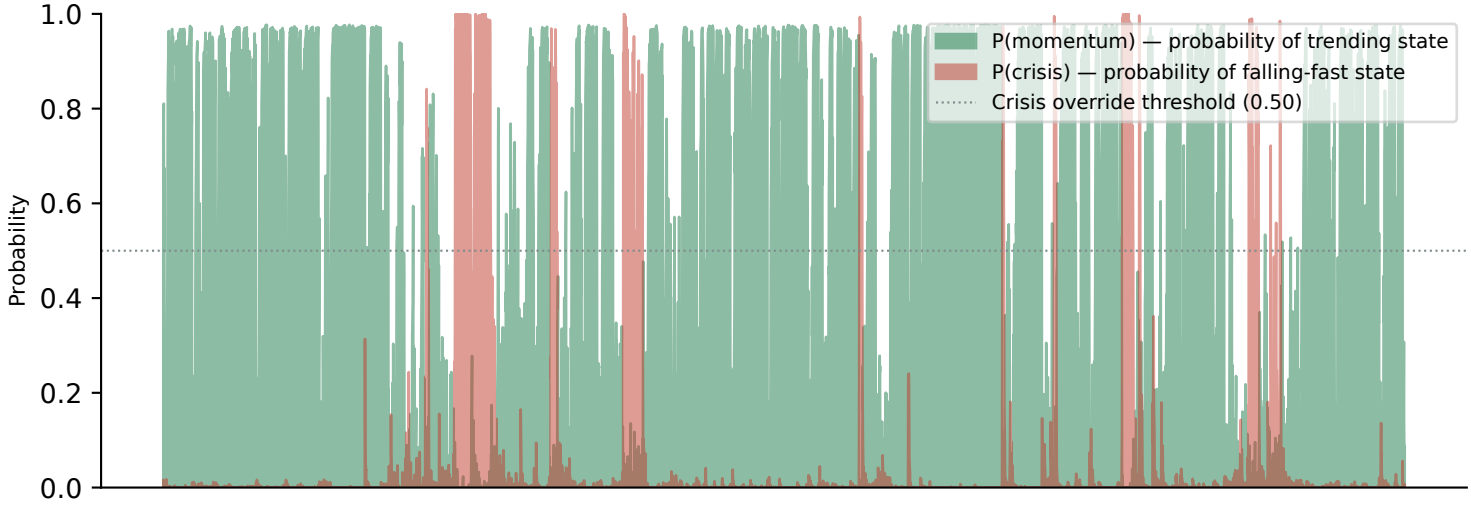
Green = momentum (hold). Red = reversion (cash). Grey = mixed (cash).
Green = momentum, red = crisis/reversion, grey = mixed. Covers dot-com, GFC 2008, COVID 2020, 2022 bear market.

Geometric Detector — Straightness Ratio (15-day window)



Near 1.0 = price moved in a straight line (momentum). Near 0.0 = price zigzagged (mean-reversion).
Thresholds are adaptive percentiles — no fixed values to hand-tune.

Markov Detector — Filtered State Probabilities (k=3 regimes)

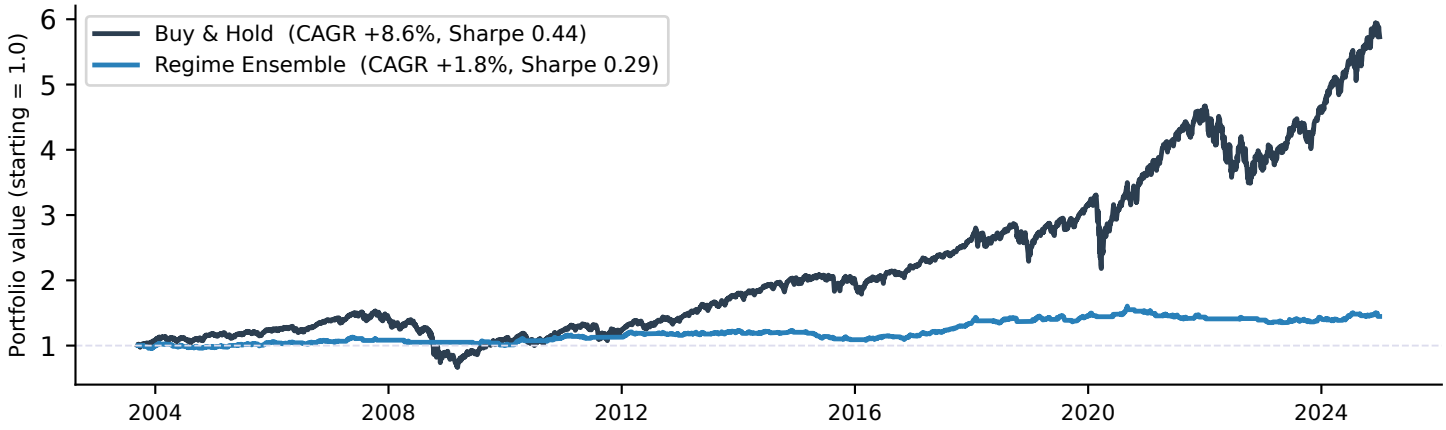


When P(crisis) crosses 0.50, buy signals are suppressed regardless of other indicators.
Filtered probabilities only — no look-ahead bias.

Results & Limitations

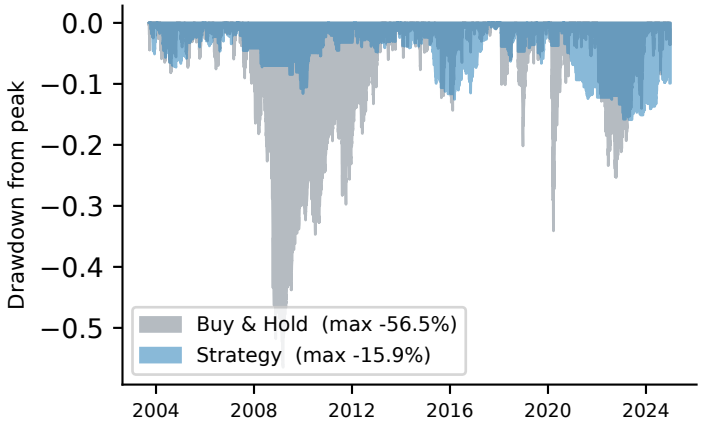
Equity curves, drawdowns, transaction cost sensitivity, and what to watch out for

Equity Curves — Strategy vs Buy & Hold (0 bps costs)



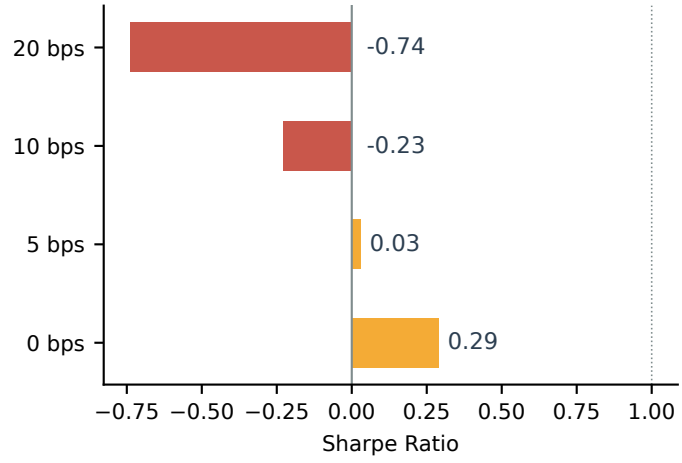
The strategy does NOT outperform buy-and-hold on raw returns (+1.8% vs +8.6% CAGR over 25 years).
The edge is risk reduction: max drawdown cut from -56.5% to -15.9%. At 5-10bps costs, returns compress further.

Drawdown Comparison



Strategy reduces drawdowns across all major crashes (dot-com, GFC, COVID, 2022).

Transaction Cost Sensitivity



~22 regime switches/year means costs add up fast.
At 20bps round-trip, strategy is unprofitable.

Key Limitations — Read Before Drawing Conclusions

⚠ **Longer sample, but single asset**
2000-2025 covers dot-com, GFC, COVID -- but it is still only SPY. Cross-validate on other tickers.

⚠ **Transaction costs are material**
At 10bps round-trip, Sharpe halves. At 20bps the strategy loses money. Manage turnover.

⚠ **In-sample threshold calibration**
Regime thresholds were tuned on the full dataset. Real-time use requires expanding-window recalibration.

⚠ **Reversion signal is weak**
Momentum signal is statistically significant (p=0.034). Reversion is not (p=0.29). Don't short on it.

⚠ **Single asset only**
Everything here is SPY. Regime structure may not generalise to other assets, sectors, or markets.